India CPI Forecasting

Project and Dataset Analysis

A source-by-source data acquisition and modeling readiness review

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Prepared from the supplied 210-page system/workflow PDF

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Scope boundary

This is a research-prototype data package, not an official RBI forecast or policy system. Where the requested official series could not be retrieved directly, the bundle labels a related substitute rather than presenting it as the same variable.

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1 Executive assessment

The PDF describes an **agentic monetary-policy and inflation simulator for India**, but its explicit priority reset is narrower: first establish one trustworthy, reproducible forecast for CPI Combined General Group inflation. The target is CPI inflation measured year over year; it is not a prediction of the index level alone. Forecast horizons are 1, 3, 6, and 12 months, with the first implementation emphasizing 1-, 3-, and 6-month model-gated forecasts.

The central architecture rule is sound: language models may retrieve, organize, critique, and explain evidence; deterministic economic and statistical tools calculate forecasts and scenario paths; the database stores source evidence and artifacts; an API serves validated results; and the interface displays provenance. The system must not imply that conditional model paths identify causal policy effects or speak for the RBI.

Practical reading of the PDF: the first acceptance milestone requires dated official CPI releases and a dated repo-rate source. The shopping-list priority inventory then names five core series: CPI, IIP, repo, Brent, and INR/USD. WPI and food indices are explicitly optional at first. The broader architecture adds macroeconomic, financial, weather, and global context only after the core series are clean.

2 What data the project actually needs

2.1 The five priority series

Series	PDF priority	Preferred source	Required handling
CPI Combined General Group	Target / core	MoSPI / NSO	Monthly index level plus release date, base year, and vintage; derive YoY from the index.
IIP General Index	Core activity feature	NSO / MoSPI	Monthly index; record vintage and revision lag; prefer first-release values in historical folds.
RBI repo rate	Core policy feature	RBI	Event-dated changes with effective date, basis-point change, and MPC date.
Brent crude	Core external feature	EIA or documented public market feed	Daily USD/barrel; calculate a documented monthly average, not an undocumented month-end value.
INR/USD reference rate	Core external feature	RBI reference-rate series	Daily INR per USD; document reference-rate basis and daily-to-monthly averaging window.

2.2 The wider feature families

The PDF's feature groups and diagrams name the following additional series or concepts. Some are source datasets; others are derived features that must be computed from source observations.

- **Inflation and prices:** food CPI/index, CPI food inflation, fuel/energy and core-inflation proxies, Wholesale Price Index (WPI), and food-price components.
- **Activity and demand:** gross domestic product / gross value added (GDP/GVA), monthly IIP growth, Purchasing Managers' Index (PMI) where licensed/available, and consumption proxies.
- **Policy and liquidity:** repo, Standing Deposit Facility (SDF), Marginal Standing Facility (MSF), Cash Reserve Ratio (CRR), liquidity conditions, and the real repo rate.
- **External/global:** INR/USD, Brent and other commodity indices, US CPI, Federal Funds rate, and global rates.
- **Financial conditions:** credit growth, broad money (M3) (M3), government-bond yields, and financial-condition measures.
- **Supply/weather:** rainfall deviations, agricultural/food-price components, and harvest shocks.
- **Derived dynamics:** CPI lags, rolling means and volatility, seasonal terms, YoY transformations, oil/FX growth, and real repo.

Derived features are not extra raw datasets. Each must retain lineage to its input series and be computed inside the forecast-time information set. For example: `CPI_YoY(t) = 100 × [CPI(t) / CPI(t−12) − 1]`; the simple real-repo proxy is nominal repo less CPI YoY. The latter is a feature construction, not a causal model.

3 Dataset bundle: scope and source decisions

The companion ZIP is organized as immutable raw downloads, a long-form World Bank source panel, and metadata/documentation. It prioritizes complete, openly retrievable history and broad context while avoiding invented values. Per-file sizes, SHA-256 hashes, and tabular row counts are in `metadata/FILE_INVENTORY.csv`; coverage, source URLs, and downloaded/proxy/missing status are in the dataset catalog, acquisition manifest, and source notes.

3.1 Primary Indian data and close substitutes

- **MoSPI/NSO CPI:** the package keeps the 2012-base history, the separately published 2024-base back series, current 2024-base detailed releases, and item/group records as distinct files. These are not concatenated into a synthetic continuous CPI series. The official base-year/revision boundary and release date must remain visible.
- **IIP:** the PDF names official MoSPI/NSO monthly releases. Where bulk source files are unavailable through the public endpoints, use a clearly labelled OECD/FRED India industrial-production series as a proxy and validate its base, definition, and coverage before modeling. A proxy is not the official release vintage.
- **Repo rate:** RBI's DBIE is the preferred source. The RBI public landing page exposes current indicators; an official workbook route may require a browser CAPTCHA. The bundle uses an OECD series mirrored by FRED, **Central Bank Rates: Total for India**, as a long-history policy-rate proxy through December 2023, and captures a separate RBI landing-page snapshot. The OECD/FRED series is **not asserted to equal the RBI repo rate** for every regime; it does not provide the missing event dates or post-2023 history. Use it only with an explicit proxy label, and replace it with an RBI event history before treating it as the policy repo series.

- **INR/USD:** use the RBI reference-rate history where retrievable; FRED exchange-rate series and World Bank annual official-exchange-rate observations are related fallbacks. Daily market/reference series and annual WDI rates are not interchangeable.
- **Brent:** the preferred bundle source is the US Energy Information Administration (EIA) daily Brent series. EIA West Texas Intermediate (WTI) is included as a separate comparator, never renamed as Brent. World Bank monthly commodity tables provide an additional cross-check.
- **WPI:** an official Office of the Economic Adviser (OEA) monthly workbook is included where download succeeds. Keep it distinct from CPI; the base is 2011–12 in the downloaded series.

3.2 Weather, global, financial, and activity extensions

- **IMD rainfall:** the regional monthly/seasonal summary covers 1901–2024; all 13 high-resolution 0.25-degree daily NetCDF files for 2012–2024 were downloaded, matching the CPI-era modeling window rather than the full slow 1901–2024 grid. The per-year manifest records file status and sizes. Use spatial aggregation to India or crop-region masks before creating monthly weather features; rainfall is not a ready-to-join national scalar.
- **World Bank WDI:** a broad all-country panel of annual indicators supplies consumer-price context, GDP/GVA and activity, exchange rates, interest-rate proxies, credit, broad money, household consumption, agricultural output, trade, population, and fiscal context. Its annual frequency is too low to stand in for monthly core regressors.
- **IMF WEO:** the all-country World Economic Outlook workbook is a broad annual macro panel containing historical estimates and projections. Keep projected values distinguishable from realized observations and from release-vintage data.
- **FRED / OECD:** the bundle includes selected US CPI, Federal Funds and market rates, bond yields, exchange rates, commodity, India/OECD industrial-production, and OECD India CPI index/YoY series as a cross-check. The IMF's separate monthly CPI dataset is documented but no raw IMF CPI extract was retrieved; the OECD series is a related alternative, not an identical IMF or MoSPI release vintage. These remain external features or documented alternatives, not substitutes for Indian official CPI/repo release vintages.
- **Private PMI:** the PDF qualifies PMI as "if licensed/available." Do not redistribute subscription-only survey files. Use open monthly IIP and, if obtained from a documented public source, high-frequency transaction/activity measures as separate proxies; annual household consumption from WDI is only a low-frequency fallback.

3.3 What remains a gap

An exact, release-aware RBI repo-event file; RBI reference-rate history if the downloaded alternate is used; first-release IIP vintages; open, license-cleared monthly PMI; official monthly food/core/fuel component history with matching base; and any licensed microdata remain gaps unless the acquisition manifest proves otherwise. Missing coverage is documented, not filled with synthetic observations. The dataset catalog marks each item as **downloaded**, **proxy**, **partial snapshot**, or **not retrieved**.

3.4 Acquisition snapshot (8 October 2026)

Collection	Contents	Size in records	Interpretation
MoSPI CPI, 2012 base	All-India group history plus item history	15,150 + 42,159	Official source API records; retain month, sector, state, base year, and status.
MoSPI CPI, 2024 base	Back series plus current detailed all-India	432 + 40,080	Separate files; do not silently splice with the 2012 base.
World Bank WDI	Selected annual indicators, all countries	612,150 records; 35 of 36 requested indicators	The row count includes API country-year rows with null values; one unavailable PPP code was not imputed.
IMD rainfall summary	5 regions × annual/ monthly/seasonal fields	620 region-year records	124 years per region, 1901–2024.
IMD daily rainfall grid	0.25-degree annual NetCDF files	13 files; 2012–2024	All target-window years succeeded; the 31.9 MiB compressed grids require spatial aggregation before use.
External data	IMF WEO; World Bank Pink Sheet; OEA WPI; EIA Brent/WTI; FRED/ OECD; OECD CPI	Multiple original workbooks and time series	See the ZIP's file inventory for exact paths, byte sizes, checksums, and tabular counts.

The RBI DBIE landing snapshot captured five current indicators on 8 October 2026: repo 5.25%, reverse repo 3.35%, CRR 3.00%, SLR 18.00%, and SDF 5.00%. The page did not specify the effective date of each rate. The separate OECD/FRED monthly central-bank-rate series ends at 6.75% in December 2023 (672 monthly observations); this is a long-history **proxy**, not an exact substitute for RBI repo events.

The bundle is sizeable in rows and source diversity, but the monthly CPI target history—not the number of external rows—sets the effective supervised-learning sample. Annual country panels and daily grids do not create additional monthly CPI labels.

4 How to use the files without invalidating the forecast

1. Start from the exact raw file and its source URL in the manifest; verify the recorded hash before analysis.
2. Keep CPI base-year vintages separate. Build the headline target from the documented index and retain the source release date.
3. Normalize source observations into long form with at least `series_id`, `ref_month`, `vintage_date`, `value`, `unit`, `source_url`, `fetched_at`, and relevant metadata such as `base_year`.
4. At a simulated forecast origin, include only observations whose `vintage_date` is on or before that origin. A later revision is available only after its release date.
5. Convert daily Brent and FX to monthly features with one declared calendar/trading-day convention; convert rainfall grids to geography-specific summaries before joining.

6. Keep frequency and definition visible: annual WDI, monthly CPI, daily financial series, quarterly GDP/GVA, and event-dated policy rates are different objects. Aligning them to month-end does not change their native frequency or release lag.
7. Flag coverage below the PDF's 95% feature gate; exclude a feature or report it as incomplete rather than silently imputing it.

The PDF's minimum dataset-level canonical fields are `series_id`, `ref_month`, `vintage_date`, `value`, `unit`, `source_url`, and `fetched_at`, plus `base_year` and source-specific metadata. Store raw files unchanged; write normalized observations to an interim layer and model-facing aligned features to a processed layer. The bundle is a collection of source downloads and documented alternatives, not a claim that the full production ingestion/model pipeline has already been validated.

5 Model and evaluation plan in the PDF

The ordered model ladder is deliberately conservative:

- **M0 — seasonal-naive:** carry forward the last YoY value; eligible after 24 months; horizons 1, 3, 6, 12.
- **M1 — drift / random walk with drift:** 36 months; the same horizons.
- **M2 — ARIMA on CPI YoY:** 60 months; the same horizons.
- **M3 — ARIMAX / ridge:** 84 months; horizons 1, 3, 6, and only release-aware features available at the origin.
- **M4 — optional VAR (CPI + IIP + repo):** 120 aligned months; horizons 1, 3, 6; add only if M3 beats M2.

Use an expanding-window expanding-window rolling-origin backtest (rolling-origin backtest) scheme. Every rung must beat the rung below it on identical folds before becoming the served default; M0 remains the honesty floor. Report per-horizon RMSE, MAE, and MASE versus M0, plus 80% interval coverage. Preserve each fold's origin, training range, predictions, metrics, fitted model, data vintage, and `run_id`. Random shuffling, future revisions, full-sample scaling, and tuning outside each fold violate the workflow's information rule.

Scenario tools may calculate baseline versus shocked paths (the PDF's examples include oil +20%, INR depreciation of 5%, food inflation +1.5 percentage points, or repo changes). These are conditional simulations under stated assumptions. Unless a separately validated causal design is added, do not say the policy shock **caused** a given inflation change.

6 Source, access, and interpretation notes

- **MoSPI / NSO:** official CPI and IIP source family; [MoSPI e-Sankhyiki](https://esankhyiki.mospi.gov.in/) and [MoSPI CPI catalogue/API](https://api.mospi.gov.in/).
- **RBI / DBIE:** official policy and macroeconomic data; [Database on Indian Economy](https://data.rbi.org.in/) and [RBI Handbook of Statistics](https://www.rbi.org.in/Scripts/AnnualPublications.aspx?head=Handbook%20of%20Statistics%20on%20Indian%20Economy).
- **EIA:** daily Brent/WTI petroleum price workbooks; [EIA petroleum data](https://www.eia.gov/dnav/pet/hist_xls/).
- **FRED / OECD:** series metadata and long-form economic series; [FRED](https://fred.stlouisfed.org/), including the [India central-bank-rate proxy](https://fred.stlouisfed.org/series/IRSTCB01INM156N), sourced from the OECD Main Economic Indicators.

- **IMD:** rainfall summaries and gridded rainfall archive; [IMD gridded rainfall](https://www.imdpune.gov.in/cmpg/Griddata/RF25.php).
- **World Bank:** [World Development Indicators](https://data.worldbank.org/) and [Commodity Markets Pink Sheet](https://www.worldbank.org/en/research/commodity-markets).
- **IMF:** [World Economic Outlook data](https://www.imf.org/en/Publications/WEO/weo-database/2026/April).
- **UPI activity proxy:** [NPCI product statistics](https://www.npci.org.in/product/upi/product-statistics) and any Dataful mirrored export are third-party/public-statistics routes; preserve original source and terms. Do not infer that UPI transaction counts are household consumption or nominal GDP.

FRED series can contain copyrighted OECD-sourced data; follow the series-note citation and reuse terms. Public availability is not a blanket redistribution license. Original provider terms govern. The data package records source URLs and retrieval timestamps; its inclusion does not certify every source's license for commercial redistribution.

7 Glossary

base year – **index base year**: The reference period used to scale an index; CPI vintages with different bases must not be silently spliced.

Brent – **Brent crude oil**: An international oil benchmark; daily USD-per-barrel observations are generally aligned to monthly averages.

conditional scenario – **conditional scenario projection**: A forecast path under stated assumptions; it is not evidence that a policy change caused the outcome.

CPI – **Consumer Price Index**: The consumer-price index level used to calculate inflation; preserve the published base year and release vintage.

GDP/GVA – **gross domestic product / gross value added**: Aggregate activity measures; available mostly at quarterly or annual frequency and released with a lag.

IIP – **Index of Industrial Production**: A monthly measure of industrial activity; the general index is the project's priority activity feature.

INR/USD – **Indian rupees per US dollar**: The exchange-rate quotation requested by the project; verify units and averaging conventions.

look-ahead leakage – **look-ahead information leakage**: Use of future observations, revisions, or fitted preprocessing information that would not have been available at the forecast origin.

M3 – **broad money (M3)**: A monetary aggregate and optional financial-condition feature; definitions and frequency must match the source.

MASE – **mean absolute scaled error**: Forecast error scaled against a naive benchmark, used alongside RMSE and MAE.

PMI – **Purchasing Managers' Index**: A survey-based activity indicator; access and licensing may restrict redistribution.

repo rate – **RBI policy repo rate**: The policy rate specified by the workflow; event/effective dates and MPC dates must be kept.

vintage – **data vintage**: The values and revisions available on a particular release date, essential for real-time backtesting.

rolling-origin backtest – **expanding-window rolling-origin backtest**: Repeated historical forecasts that use only observations released by each simulated forecast origin.

WPI – **Wholesale Price Index**: A complementary price indicator, useful for producer/input-price context but not a replacement for CPI.

YoY – **year over year**: Percentage change from the same month one year earlier; the project's headline target is CPI YoY, not the CPI index level.